
NASH BARGAINING PREFERENCE OPTIMIZATION

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Paper under double-blind review

ABSTRACT

Aligning one large language model (LLM) to several objectives requires two decisions: how to represent preferences within each objective and how to choose one compromise across objectives. The first decision is difficult because pairwise judgments can be cyclic in realistic settings. For a medical-advice prompt, annotators may prefer a concise answer to a verbose one, the verbose answer to an incomplete one, yet the incomplete answer to the concise one because it includes a critical safety caveat. No scalar ranking can satisfy all three comparisons. Nash Learning from Human Feedback avoids this restriction by treating pairwise preferences as game payoffs, but its multi-objective extensions still require an aggregation rule. We introduce Nash Bargaining Preference Optimization (NBPO), which evaluates each objective through a regularized preference game and applies Nash bargaining as an aggregation rule to its improvement over a common reference policy. Our main technical contribution is a practical optimizer for these policy-dependent game values: a constrained KL-proximal update handles zero surplus at the reference, closed-form opponent reweighting exposes weak responses, and pairwise log-ratio regression removes the intractable policy normalizer while learning directly from binary comparisons. Under exact population updates, the final iterate achieves convergence with rate $O(1/T)$. Experiments compare preference-based and reward-model-based multi-objective baselines under independent evaluation, while controlled rescaling tests verify the predicted invariance.

1 INTRODUCTION

Reinforcement Learning from Human Feedback (RLHF) is a standard approach for aligning large language models (LLMs) with human judgments (Christiano et al., 2017; Ouyang et al., 2022). Standard RLHF learns a scalar reward from pairwise comparisons. Direct Preference Optimization removes the separately trained reward model, but still assumes that comparisons are generated by an implicit scalar reward through a Bradley–Terry (BT) model (Rafailov et al., 2023; Bradley & Terry, 1952). Both approaches therefore require every response to admit one scalar score whose ordering explains all pairwise preferences.

This assumption can fail even within a single alignment objective. Human judgments often reflect several latent attributes whose relative importance changes with the prompt and the pair being compared. Consider three answers to a medical-advice question. Response A is concise and actionable but omits an important caveat; response B is comprehensive but verbose; and response C is less complete but includes the critical safety warning. Annotators may prefer A to B for relevance, B to C for completeness, and C to A for safety and reliability. These comparisons are cyclic: $A \succ B$, $B \succ C$, and $C \succ A$. Fitting the first two comparisons with a scalar reward requires $r(A) > r(B) > r(C)$, which contradicts the third. As a result, optimizing the fitted scalar reward can favor a response that appears best under the learned ranking but is still beaten by another response according to the original pairwise judgments.

Nash Learning from Human Feedback (NLHF) avoids this misspecification by treating prompt-conditioned comparisons directly as payoffs in a zero-sum preference game, and INPO, SPPO, and related methods provide scalable equilibrium-seeking updates (Munos et al., 2024; Zhang et al., 2025; Wu et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Extending this view to multiple objectives introduces a second problem: after representing each objective without imposing transitivity, which compromise should one deployed policy realize? Scalar multi-objective methods expose this trade-off through weights, priorities, constraints, policy interpolation,

or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b; Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). Recent game-theoretic methods preserve general pairwise feedback but combine objectives through Blackwell target sets or a worst-objective criterion (Blackwell, 1956; Bhatia et al., 2020; Zhang et al., 2026).

Classical Nash bargaining offers a priority-free way to select one compromise. It maximizes the product of each objective’s improvement over a common fallback (Nash, 1950). In alignment, the reference policy provides a natural fallback: an aligned policy should improve each objective relative to the model that would otherwise be deployed. The classical Nash solution is Pareto optimal and invariant to independent changes in the units used for different objectives. Prior alignment work has applied Nash welfare to learned scalar rewards, while game-based preference methods have focused mainly on minimax or Blackwell solutions (Zhong et al., 2024a; Navon et al., 2022; Zhang et al., 2026). What remains is a practical method for Nash bargaining when every objective value is itself defined by a policy-dependent preference game rather than by a fixed scalar reward.

We introduce Nash Bargaining Preference Optimization (NBPO) for this setting. Within each objective, NBPO evaluates the learner against an adaptive opponent that emphasizes responses exposing the learner’s weaknesses, while a KL penalty keeps the opponent near the reference policy. Across objectives, it maximizes the Nash product of the resulting improvements over that same reference. The standard bargaining properties are useful in LLM alignment because they ensure that no objective falls below the fallback at the population optimum, no other feasible policy can improve one objective without worsening another, and arbitrary judge-specific units do not alter the selected compromise.

The main technical challenge is optimization, not re-deriving the classical bargaining axioms. Three issues arise simultaneously: the reference policy has zero improvement and lies on the boundary of the logarithmic objective; each adaptive opponent depends on the policy being optimized; and the ideal exponential policy update contains an intractable normalizing constant. We address these issues with a constrained KL-proximal formulation, closed-form opponent reweighting, a low-dimensional dual update whose weights are inversely proportional to achieved improvements, and partition-free pairwise log-ratio regression from binary comparisons.

Our contributions are:

- We formulate multi-objective alignment as Nash bargaining over objective-wise regularized preference-game values, thereby separating within-objective preference representation from across-objective compromise selection.
- We derive a practical training procedure that handles the zero-surplus reference boundary, alternates closed-form opponent updates with policy fitting, and learns the policy update without fitting scalar rewards or evaluating a language-model partition function.
- We establish $O(1/T)$ last-iterate convergence for exact population proximal updates and evaluate the method against matched preference-optimization controls and reward-model-based multi-objective baselines under independent held-out metrics.

2 BACKGROUND

General preference models. We use contextual-bandit notation. A prompt (context) x is drawn from a distribution $\mathcal{D}$, a response y is an action from a finite set $\mathcal{Y}_x$, and $\pi(y | x)$ is the policy. The reference policy is denoted by π_{ref} . For the finite-dimensional population analysis, the support of $\mathcal{D}$ and every $\mathcal{Y}_x$ are finite, and

$$\Pi = \prod_{x \in \text{supp}(\mathcal{D})} \Delta(\mathcal{Y}_x)$$

is the contextual policy simplex. Objective $k \in [K]$ is represented by a pairwise preference model $P_k(y \succ z | x) \in [0, 1]$ satisfying

$$P_k(y \succ z | x) + P_k(z \succ y | x) = 1, \quad P_k(y \succ y | x) = \frac{1}{2}. \quad (1)$$

We center each comparison at a tie,

$$\Delta_k(y, z | x) = P_k(y \succ z | x) - \frac{1}{2}, \quad \Delta_k(y, z | x) = -\Delta_k(z, y | x), \quad (2)$$

and define the expected objective- k payoff of learner policy π against comparator policy ν as

$$J_k(\pi, \nu) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot | x), z \sim \nu(\cdot | x)} \Delta_k(y, z | x). \quad (3)$$

Thus $J_k(\pi, \nu) > 0$ means that π is preferred to ν on average under objective k . This representation does not assume a scalar reward or transitive preferences.

Bargaining problems. A K -dimensional bargaining problem consists of a feasible value set $\mathcal{G} \subseteq \mathbb{R}^K$ and a disagreement point $d \in \mathcal{G}$, the outcome obtained if no agreement is reached. For a feasible value vector v , define its objective-wise surplus over disagreement as

$$s_k(v) := v_k - d_k, \quad s(v) = (s_1(v), \dots, s_K(v)). \quad (4)$$

A value vector is individually rational when $s(v) \geq 0$ componentwise, and strictly individually rational when $s(v) > 0$. On a compact, convex, and downward-closed feasible set with at least one strictly individually rational point, the classical Nash solution is

$$v^N \in \arg \max_{v \in \mathcal{G}: s(v) > 0} \sum_{k=1}^K \log s_k(v). \quad (5)$$

Equivalently, it maximizes the product of the objective-wise surpluses (Nash, 1950).

3 OBJECTIVE-WISE PREFERENCE GAMES

We first convert each objective's pairwise preference model into a scalar policy value without imposing a scalar reward on individual responses. For objective k , an adaptive comparator ν searches for responses against which the learner performs poorly. The reference policy π_{ref} anchors this comparator and later serves as the fallback in the bargaining problem.

Definition 1 (Entropy-regularized game value). *For objective k and temperature $\beta_k > 0$, define*

$$V_{k, \beta_k}(\pi) = \min_{\nu} \{J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x))\}. \quad (6)$$

The comparator minimizes the learner's pairwise payoff, so it emphasizes responses that expose weaknesses of the current policy. The KL term prevents it from moving arbitrarily far from π_{ref} . Accordingly, a large β_k produces a local evaluation near the reference, whereas $\beta_k \downarrow 0$ approaches the unregularized worst-comparator value.

For a candidate comparator response z , let

$$m_k^\pi(z | x) := \mathbb{E}_{y \sim \pi(\cdot | x)} \Delta_k(y, z | x) \quad (7)$$

be the learner's expected margin against z . Smaller values identify stronger challenges. The inner problem in (6) has the closed-form solution

$$\nu_{k, \pi}^*(z | x) = \frac{\pi_{\text{ref}}(z | x) \exp\{-m_k^\pi(z | x)/\beta_k\}}{\mathbb{E}_{z' \sim \pi_{\text{ref}}(\cdot | x)} \exp\{-m_k^\pi(z' | x)/\beta_k\}}, \quad (8)$$

$$V_{k, \beta_k}(\pi) = -\beta_k \mathbb{E}_{x \sim \mathcal{D}} \log \mathbb{E}_{z \sim \nu_{k, \pi}^*(\cdot | x)} \exp\{-m_k^\pi(z | x)/\beta_k\}. \quad (9)$$

Thus the opponent is obtained by reweighting reference responses toward those with smaller margins, and the game value is a soft minimum of these margins.

Proposition 1 (Game-value structure). *For every k and $\beta_k \geq 0$, V_{k, β_k} is continuous and concave in π . For $\beta_k > 0$, its policy-gradient coordinate is*

$$[\nabla_{\pi} V_{k, \beta_k}(\pi)]_{x, y} = \mathcal{D}(x) Q_k^\pi(y | x), \quad Q_k^\pi(y | x) := \mathbb{E}_{z \sim \nu_{k, \pi}^*(\cdot | x)} \Delta_k(y, z | x). \quad (10)$$

The proof is in Appendix A.1. The response-level score $Q_k^\pi(y | x)$ is the expected margin of response y against the current objective- k opponent; it therefore measures the local effect of increasing $\pi(y | x)$. An adaptive opponent can distinguish policies that have the same win rate against one fixed reference but different vulnerabilities to other responses. Appendix A.2 gives a finite cyclic example at the unregularized endpoint $\beta = 0$; this illustration is not used in the later assumptions or proofs.

4 NASH BARGAINING OVER GAME-VALUE IMPROVEMENTS

We now combine the objective-wise game values using the reference policy as the common fallback. For each objective,

$$d_k := V_{k,\beta_k}(\pi_{\text{ref}}), \quad s_k(\pi) := V_{k,\beta_k}(\pi) - d_k. \quad (11)$$

The surplus $s_k(\pi)$ is the improvement of π over the deployed reference under objective k .

Assumption 1 (Joint improvement and support). *The reference π_{ref} has full support, and there exists a policy $\bar{\pi}$ such that $s_k(\bar{\pi}) > 0$ for every k .*

The assumption states that at least one policy improves all objectives over the fallback. Under this condition, the NBPO population target is

$$\pi^N \in \arg \max_{\pi \in \Pi_+} F(\pi), \quad F(\pi) := \sum_{k=1}^K \log s_k(\pi), \quad \Pi_+ := \{\pi \in \Pi : s_k(\pi) > 0, \forall k\}. \quad (12)$$

Equivalently, introducing auxiliary lower bounds $u_k > 0$ yields

$$(\pi^N, u^N) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \sum_{k=1}^K \log u_k \quad \text{s.t.} \quad u_k \leq s_k(\pi), \quad k \in [K], \quad \pi \in \Pi. \quad (13)$$

Because the objective is increasing in every u_k , each constraint is tight at an optimum and $u_k^N = s_k(\pi^N)$.

To connect (12) to the classical bargaining problem, define

$$V_\beta(\pi) = (V_{1,\beta_1}(\pi), \dots, V_{K,\beta_K}(\pi)), \quad d = (d_1, \dots, d_K),$$

and the downward-closed feasible value set

$$\mathcal{G}_\beta := \{v \in \mathbb{R}^K : d \leq v \leq V_\beta(\pi) \text{ componentwise for some } \pi \in \Pi\}. \quad (14)$$

Proposition 2 (Policy-induced Nash bargaining). *Under Assumption 1, $\mathcal{G}_\beta$ is compact, convex, downward closed above d , and nondegenerate. The policies solving (12) realize the classical Nash solution of $(\mathcal{G}_\beta, d)$. Consequently, every optimizer has positive surplus in every objective and is Pareto optimal in game-value space; all optimizers induce the same game-value vector; and the optimizer set is unchanged by independent transformations*

$$(V_{k,\beta_k}, d_k) \mapsto (c_k V_{k,\beta_k} + b_k, c_k d_k + b_k), \quad c_k > 0.$$

Appendix A.3 verifies the feasible-set conditions. Once those conditions hold, the stated outcome properties are standard consequences of the classical Nash bargaining solution (Nash, 1950); we do not claim them as new bargaining axioms. In LLM alignment, they mean that the population solution improves every objective over the reference, cannot be Pareto dominated by another feasible policy, and does not change merely because different judges report their values in different numerical units. Appendix A.4 compares these guarantees with utilitarian and maxmin aggregation.

5 NASH BARGAINING PREFERENCE OPTIMIZATION

Optimizing (12) presents three practical difficulties. First, the reference has $s_k(\pi_{\text{ref}}) = 0$, so the logarithmic objective and its gradient are undefined there. Second, each objective's opponent depends on the policy being optimized. Third, the ideal exponential policy update contains a language-model normalizer that cannot be computed. This section addresses the three issues with a constrained proximal step, alternating opponent and policy updates, and pairwise log-ratio regression.

5.1 CONSTRAINED PROXIMAL UPDATE AND ADAPTIVE OBJECTIVE WEIGHTS

For $\pi \in \Pi_+$, Proposition 1 and the chain rule give

$$[\nabla_\pi F(\pi)]_{x,y} = \mathcal{D}(x) \sum_{k=1}^K \frac{Q_k^\pi(y | x)}{s_k(\pi)}. \quad (15)$$

The inverse-surplus factors in (15) emphasize objectives that have improved less. Because this expression cannot be evaluated at π_{ref} , outer stage t instead solves the constrained KL-proximal problem

$$\begin{aligned} (\pi_{t+1}, u_{t+1}) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \quad & \sum_{k=1}^K \log u_k - \frac{1}{\eta_t} D(\pi \| \pi_t) \\ \text{s.t.} \quad & u_k \leq s_k(\pi), \quad k \in [K], \\ & \pi \in \Pi, \end{aligned} \quad (16)$$

where $\pi_0 = \pi_{\text{ref}}$, $\eta_t > 0$, and $D(\pi \| \pi_t) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\pi(\cdot | x) \| \pi_t(\cdot | x))$. Assumption 1 supplies a feasible positive-surplus policy, while π_{ref} serves only as the initial proximal center.

The dual of (16) is

$$\min_{\lambda \in \mathbb{R}_{++}^K} \phi_t(\lambda), \quad \phi_t(\lambda) = - \sum_{k=1}^K \log \lambda_k - K + \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (17)$$

For fixed λ , denote the unique inner maximizer by

$$\pi_t^*(\lambda) := \arg \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (18)$$

Danskin's theorem yields the low-dimensional dual gradient

$$\nabla_k \phi_t(\lambda) = s_k(\pi_t^*(\lambda)) - \frac{1}{\lambda_k}. \quad (19)$$

The complete dual derivation is in Appendix A.5.

Proposition 3 (Inverse-surplus objective weights). *At an optimal primal–dual solution of (16),*

$$u_{t+1,k} = s_k(\pi_{t+1}), \quad \lambda_{t+1,k} = \frac{1}{u_{t+1,k}} = \frac{1}{s_k(\pi_{t+1})}. \quad (20)$$

Thus the constrained problem recovers the same inverse-surplus weighting visible in (15): objectives with smaller achieved improvements receive larger weights, without choosing those weights in advance.

5.2 SAMPLED POLICY FITTING AND DUAL UPDATE

Fix outer stage t and a dual vector λ . Because the opponent $\nu_{k,\pi}^*$ depends on the learner, we approximately solve (18) by alternating two operations. At inner iteration r , first construct $\nu_{k,\pi(r)}^*$ for each objective. Then hold these opponents fixed and fit the next policy. Define

$$Q_k^{(r)}(y | x) := \mathbb{E}_{z \sim \nu_{k,\pi(r)}^*(\cdot | x)} \Delta_k(y, z | x).$$

The ideal fixed-opponent update is

$$\pi^{(r+1)}(y | x) \propto \pi_t(y | x) \exp \left\{ \eta_t \sum_{k=1}^K \lambda_k Q_k^{(r)}(y | x) \right\}. \quad (21)$$

Its normalizer disappears when comparing two responses. Let

$$h_t(\pi; x, y, y') := \log \frac{\pi(y | x)}{\pi_t(y | x)} - \log \frac{\pi(y' | x)}{\pi_t(y' | x)}. \quad (22)$$

Then (21) implies

$$h_t(\pi^{(r+1)}; x, y, y') = \eta_t \sum_{k=1}^K \lambda_k \left(Q_k^{(r)}(y | x) - Q_k^{(r)}(y' | x) \right). \quad (23)$$

Algorithm 1 Nash Bargaining Preference Optimization

Require: Reference π_{ref} , preference models $P_{1:K}$, temperatures $\beta_{1:K}$, outer stages T , dual steps M , policy-opponent steps R .

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1: Estimate  $d_k = V_{k, \beta_k}(\pi_{\text{ref}})$  and set  $\pi_0 \leftarrow \pi_{\text{ref}}$ .
2: for  $t = 0, \dots, T - 1$  do
3:   Initialize positive weights  $\lambda^{(0)}$  (warm-start from the previous stage when available).
4:   for  $m = 0, \dots, M - 1$  do
5:      $\hat{\pi}^{(m)} \leftarrow \text{POLICYFIT}(\pi_t, \lambda^{(m)}, R)$  using opponent reweighting and (26).
6:     Estimate  $\hat{s}(\hat{\pi}^{(m)})$  and update  $\lambda^{(m+1)}$  by (27).
7:   end for
8:    $\tilde{\pi}_{t+1} \leftarrow \text{POLICYFIT}(\pi_t, \lambda^{(M)}, R)$ .
9:   Set  $\pi_{t+1} \leftarrow \tilde{\pi}_{t+1}$  if all held-out empirical surpluses are positive; otherwise retain  $\pi_t$ .
10: end for
11: return  $\pi_T$ .
```

To estimate the right-hand side, sample (y, y') from a full-support pair proposal $\xi(\cdot, \cdot | x)$ and sample $z_k \sim \nu_{k, \pi^{(r)}}^*(\cdot | x)$. Query independent binary judgments

$$B_k \sim \text{Bernoulli}(P_k(y \succ z_k | x)), \quad B'_k \sim \text{Bernoulli}(P_k(y' \succ z_k | x)),$$

and set

$$Z_k^{(r)} := B_k - B'_k. \quad (24)$$

Because

$$\mathbb{E}[Z_k^{(r)} | x, y, y'] = Q_k^{(r)}(y | x) - Q_k^{(r)}(y' | x), \quad (25)$$

the weighted binary label is an unbiased target for the desired log-ratio. We therefore fit

$$\mathcal{L}_{\text{reg}}^{(t, \lambda, r)}(\pi) = \mathbb{E} \left[h_t(\pi; x, y, y') - \eta_t \sum_{k=1}^K \lambda_k Z_k^{(r)} \right]^2, \quad (26)$$

where the expectation follows the sampling process above. Recomputing the opponents after each policy fit produces a fixed point that satisfies the optimality condition of (18); Appendix A.5 gives the population argument.

After approximately solving the weighted policy problem, we estimate its surplus vector and update the positive dual weights by

$$\lambda^{(m+1)} = \Pi_{\Lambda} \left[\lambda^{(m)} - \gamma_m \left(\hat{s}(\pi_t^*(\lambda^{(m)})) - (\lambda^{(m)})^{-1} \right) \right], \quad (27)$$

where the inverse is elementwise and Π_{Λ} keeps the numerical weights in a positive bounded box. The indices have distinct roles: r alternates opponent construction and policy fitting for fixed λ ; m updates the dual weights within one proximal stage; and t commits the next outer policy.

The held-out check is an implementation safeguard rather than a finite-sample guarantee. The convergence result below concerns exact population solutions of (16).

5.3 LAST-ITERATE CONVERGENCE

Because Algorithm 1 returns the final outer policy rather than an average of past policies, we analyze the last population iterate.

Theorem 1 (Last-iterate convergence). *Under Assumption 1, let π^* maximize (12), set $\pi_0 = \pi_{\text{ref}}$, and solve (16) exactly at every outer iteration with arbitrary $\eta_t > 0$. Then $F(\pi_{t+1}) \geq F(\pi_t)$ for $t \geq 1$, and for every $T \geq 1$,*

$$F(\pi^*) - F(\pi_T) \leq \frac{D(\pi^* \| \pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (28)$$

A constant proximal step therefore gives an $O(1/T)$ final-iterate Nash-welfare gap. Appendix A.6 gives the proof.

6 EXPERIMENTS

Setup. We evaluate fixed objective panels on ULTRAFEDBACK, TL;DR, and SAFERLHF. The primary backbone and reference are Llama-3.1-8B-Instruct, with a Qwen3-8B replication. A prompted Qwen3-32B model supplies objective-specific training comparisons, queried in both presentation orders. Held-out evaluation uses models that supply none of the training labels. Appendix C reports data splits, generation, judging, optimization, and compute; Appendix H gives the objective rubrics and judge prompts.

Baselines and evaluation. We include two baseline families. Preference-based baselines are PROSPER/MaxEntBW, equal-weight utilitarian and maxmin controls over the same game-valued objectives, Uniform-INPO, MOPO, and single-objective HT-MNPO. Reward-model-based baselines are MaxMin-RLHF, Rewarded Soups, and Panacea (Chakraborty et al., 2024; Ramé et al., 2023; Zhong et al., 2024b). For methods that produce a family of policies, one operating point must be selected using validation data only. Appendix D records the reward models, policy optimizer, selection rule, and compute required for these runs.

Our primary cross-method metric is independent objective-wise win rate against the common reference,

$$W_k^{\text{eval}}(\pi) := \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot|x), y_{\text{ref}} \sim \pi_{\text{ref}}(\cdot|x)} P_k^{\text{eval}}(y \succ y_{\text{ref}} | x). \quad (29)$$

All methods are ranked by the same held-out evaluators, so this metric does not privilege the NBPO training objective. Table 2 reports the worst and average W_k^{eval} across each panel, with every objective listed in Appendix B. The adaptive game-value surpluses s_k remain useful for diagnosing whether an implementation realizes the NBPO population target, but we report them only as a secondary method-aligned diagnostic in the appendix. Table 1 evaluates general capability.

Main multi-objective comparison. Table 2 reserves the primary comparison under common held-out win rates. These entries should be filled from a prompt-disjoint evaluation set and reported as mean $\pm$ standard deviation across independent training seeds. This table replaces adaptive game surplus as the headline cross-method metric; the existing surplus results are retained in Appendix B as a diagnostic of the NBPO objective.

Separating representation and aggregation. Table 3 holds one design choice fixed at a time. Comparing the first two rows keeps Nash aggregation fixed and changes how each objective is evaluated; comparing the adaptive-game rows keeps the objective representation fixed and changes the aggregation rule. The current average reference-win estimates slightly favor the simpler fixed-reference Nash variant, so they do not yet establish an empirical benefit from adaptive opponents. The pending objective-wise columns are necessary to determine whether either method sacrifices a particular objective.

Scale invariance and optimization diagnostics. Because affine invariance is a standard Nash-bargaining property rather than a new alignment theorem, Table 4 checks whether the implemented update preserves it numerically. For each frozen response pair, we compare the sign of the aggregation-induced policy-update score before and after independently rescaling the objective coordinates. NBPO changes this direction for only 0.01% of pairs; alternative fixed-weight or maxmin rules change more often. Appendix F reports temperature, comparator-sampling, fixed-point, dual, and regression diagnostics needed to identify the remaining gap between the population objective and the neural implementation.

7 RELATED WORK

General pairwise preferences. NLHF and its scalable variants optimize directly against general pairwise preference models rather than fitting scalar rewards (Munos et al., 2024; Wu et al., 2025; Zhang et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Blackwell-style methods extend this view to vector-valued feedback and target sets (Blackwell, 1956; Bhatia et al., 2020). PROSPER introduces a maximum-entropy Blackwell winner with a scalable single-policy update and an inner worst-case choice over objective weights and opponents (Zhang et al., 2026).

Table 1: General-capability evaluation after multi-objective training. The first four columns are held-out win rates against the corresponding reference policy; IFEval and TruthfulQA use their official metrics. Reward-model baselines are pending and left blank. Higher is better.

method	Arena-Hard v2	AlpacaEval 2	Alpaca LC	MT-Bench	IFEval	TruthfulQA
<i>Backbone and reference policy: Llama-3.1-8B-Instruct</i>						
Base = $\pi_{\text{ref}} = \pi_0$	0.500	0.500	0.500	0.500	0.762	0.541
NBPO (ours)	0.558	0.554	0.554	0.553	0.701	0.545
audited code, $\eta_t=1, 300$	0.517	0.525	0.524	0.506	0.756	0.542
audited code, $\eta_t=1, 300$, filtered pairs	0.498	0.521	0.510	0.512	0.745	0.546
audited code, $\eta_t=1, 600$, filtered pairs	0.513	0.528	0.518	0.534	0.747	0.550
audited code, $\eta_t \times 8, 300$	0.527	0.538	0.532	0.541	0.732	0.545
PROSPER/MaxEntBW	0.515	0.546	0.541	0.560	0.630	0.543
Game-utilitarian	0.522	0.544	0.548	0.516	0.617	0.544
Game-absolute-maxmin	0.532	0.536	0.548	0.525	0.638	0.544
Game-surplus-maxmin	0.505	0.532	0.524	0.503	0.636	0.548
Uniform-INPO	0.527	0.517	0.535	0.503	0.645	0.541
MOPO	0.508	0.512	0.516	0.513	0.749	0.542
HT-MNPO (honesty)	0.549	0.528	0.522	0.519	0.717	0.544
HT-MNPO (truthfulness)	0.536	0.531	0.517	0.525	0.675	0.547
HT-MNPO (helpfulness)	0.530	0.550	0.546	0.494	0.649	0.542
HT-MNPO (instr. following)	0.515	0.529	0.533	0.541	0.739	0.541
MaxMin-RLHF	0.534	0.522	0.506	0.547	0.723	0.546
Rewarded Soups	0.534	0.540	0.515	0.497	0.741	0.544
Panacea	0.579	0.540	0.518	0.506	0.739	0.551
<i>Backbone and reference policy: Qwen3-8B, reasoning trace suppressed</i>						
Base = $\pi_{\text{ref}} = \pi_0$	0.500	0.500	0.500	0.500	0.815	0.544
NBPO (ours)	0.512	0.562	0.534	0.529	0.810	0.546
PROSPER/MaxEntBW	0.505	0.535	0.519	0.497	0.808	0.544
Game-utilitarian	0.509	0.528	0.511	0.497	0.810	0.545
Game-absolute-maxmin	0.510	0.531	0.514	0.503	0.806	0.545
Game-surplus-maxmin	0.500	0.534	0.523	0.516	0.804	0.544
Uniform-INPO	0.491	0.498	0.496	0.481	0.821	0.545
MOPO	0.498	0.499	0.497	0.468	0.815	0.543
HT-MNPO (honesty)	0.493	0.500	0.503	0.513	0.815	0.544
HT-MNPO (truthfulness)	0.503	0.517	0.508	0.528	0.813	0.544
HT-MNPO (helpfulness)	0.491	0.497	0.498	0.519	0.811	0.545
HT-MNPO (instr. following)	0.491	0.491	0.490	0.509	0.819	0.544
MaxMin-RLHF	0.500	0.457	0.491	0.532	0.824	0.544
Rewarded Soups	0.499	0.475	0.484	0.550	0.813	0.544
Panacea	0.478	0.469	0.485	0.519	0.826	0.543

Table 2: Primary cross-method evaluation under independent objective-wise win rates against the common reference. We report the worst and average objective win rate in each panel. Blank cells denote pending prompt-disjoint, multi-seed runs. Higher is better.

method	ULTRAFEEDBACK		TL;DR		SAFERLHF	
	worst W^{eval}	avg. W^{eval}	worst W^{eval}	avg. W^{eval}	worst W^{eval}	avg. W^{eval}
Base = π_{ref}	0.500	0.500	0.500	0.500	0.500	0.500
NBPO (ours)	0.524 $\pm$ 0.007	0.535 $\pm$ 0.006	0.528 $\pm$ 0.011	0.558 $\pm$ 0.007	0.696 $\pm$ 0.013	0.718 $\pm$ 0.012
PROSPER/MaxEntBW	0.496 $\pm$ 0.008	0.541 $\pm$ 0.007	0.444 $\pm$ 0.015	0.592 $\pm$ 0.007	0.760 $\pm$ 0.013	0.777 $\pm$ 0.012
Game-utilitarian	0.467 $\pm$ 0.008	0.500 $\pm$ 0.007	0.415 $\pm$ 0.016	0.595 $\pm$ 0.007	0.780 $\pm$ 0.012	0.787$\pm$0.012
Game-absolute-maxmin	0.459 $\pm$ 0.008	0.499 $\pm$ 0.008	0.365 $\pm$ 0.014	0.557 $\pm$ 0.006	0.781$\pm$0.012	0.782 $\pm$ 0.012
Game-surplus-maxmin	0.502 $\pm$ 0.009	0.549 $\pm$ 0.007	0.491 $\pm$ 0.013	0.602 $\pm$ 0.007	0.778 $\pm$ 0.012	0.780 $\pm$ 0.012
Uniform-INPO	0.540 $\pm$ 0.009	0.580$\pm$0.007	0.228 $\pm$ 0.015	0.519 $\pm$ 0.007	0.447 $\pm$ 0.016	0.471 $\pm$ 0.012
MOPO	0.507 $\pm$ 0.007	0.511 $\pm$ 0.007	0.513 $\pm$ 0.012	0.546 $\pm$ 0.007	0.720 $\pm$ 0.013	0.730 $\pm$ 0.012
HT-MNPO, best corner	0.528 $\pm$ 0.008	0.547 $\pm$ 0.007	0.502 $\pm$ 0.011	0.517 $\pm$ 0.008	0.470 $\pm$ 0.010	0.542 $\pm$ 0.010
MaxMin-RLHF	0.537 $\pm$ 0.007	0.557 $\pm$ 0.006	0.346 $\pm$ 0.014	0.564 $\pm$ 0.007	0.705 $\pm$ 0.014	0.736 $\pm$ 0.012
Rewarded Soups	0.534 $\pm$ 0.008	0.557 $\pm$ 0.007	0.506 $\pm$ 0.014	0.569 $\pm$ 0.007	0.728 $\pm$ 0.013	0.742 $\pm$ 0.011
Panacea	0.554$\pm$0.007	0.572 $\pm$ 0.007	0.560$\pm$0.012	0.602$\pm$0.007	0.703 $\pm$ 0.013	0.726 $\pm$ 0.011

NBPO instead keeps one regularized game value per objective and applies Nash bargaining across the improvements over a shared fallback.

Reward-based multi-objective alignment. MODPO, Rewarded Soups, and Panacea expose user-specified trade-offs through scalarization, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b). MaxMin-RLHF and RMOD optimize weakest-objective criteria, while MOPO designates a primary objective and lower-bound constraints for

Table 3: Matched component ablation on the same held-out ULTRAFEEDBACK prompts. All variants share the response pool, training budget, and evaluator. Worst-objective values are pending; existing average reference-win rates are retained. Higher is better.

variant	objective value	aggregation	worst W^{eval}	avg. W^{eval}
NBPO	adaptive game	Nash	0.509	0.556
Fixed-reference NBPO	fixed reference	Nash	0.503	0.561
Game-utilitarian	adaptive game	utilitarian	0.503	0.452
Game-surplus-maxmin	adaptive game	surplus maxmin	0.506	0.494

Table 4: Pairwise update-score sign flips (%) under three independent objective rescalings on a frozen ULTRAFEEDBACK pool. Lower is better; zero is exact invariance.

selection rule	$c^{(A)}$	$c^{(B)}$	$c^{(C)}$	mean
NBPO (ours)	0.01	0.01	0.00	0.01
Utilitarian	5.66	6.31	5.85	5.94
Uniform-INPO	5.66	6.31	5.85	5.94
MOPO	5.66	6.31	5.85	5.94
PROSPER/MaxEntBW	7.35	6.63	6.71	6.90
Surplus maxmin	16.02	13.57	0.00	9.86
Absolute maxmin	16.32	0.00	16.32	10.88

the others (Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). These methods are important empirical baselines because they explicitly fit or use scalar objective rewards, whereas NBPO learns from general pairwise feedback without assuming that each objective is transitive.

Social choice and bargaining. The Nash solution and its Pareto, uniqueness, and affine-invariance properties are classical (Nash, 1950). Social-choice work argues that alignment should state its aggregation rule explicitly (Conitzer et al., 2024). Multi-party RLHF studies utilitarian, Nash, and leximin welfare over party-specific learned rewards and separately considers reward-free von Neumann solutions (Zhong et al., 2024a); Nash bargaining has also been used to combine task gradients in multi-task learning (Navon et al., 2022). Our contribution is not a new bargaining axiom, but a formulation and practical optimizer for applying the classical Nash rule to policy-dependent objective-wise preference-game values.

8 CONCLUSION AND LIMITATIONS

NBPO separates two decisions in multi-objective preference alignment: representing possibly cyclic comparisons within each objective and selecting one compromise across objectives. It uses regularized preference games for the first decision and the classical Nash bargaining rule for the second. The bargaining guarantees are inherited once the policy-induced game-value set satisfies the standard conditions; our main contribution is the constrained and partition-free optimization procedure that makes this construction trainable from binary comparisons. The exact population proximal iteration achieves an $O(1/T)$ last-iterate Nash-welfare gap.

Several limitations remain. The current component ablation does not establish an empirical benefit from adaptive game values over the fixed-reference Nash variant. The neural algorithm also approximates the policy-opponent fixed point, dual solution, and regression target from finite samples, whereas the convergence theorem assumes exact population updates. Finally, the headline cross-method comparison must be completed under common independent metrics, reward-model baselines, and multiple training seeds. The joint-improvement assumption can also fail when no policy improves every objective over the chosen reference; in that case a positive-surplus Nash solution does not exist. Improvements in game value should not be interpreted as per-prompt safety guarantees.

AI USE STATEMENT

In this work, generative AI tools were used to critique the problem formulation, assist with mathematical derivations and proof presentation, propose experimental checks, and edit manuscript text. The authors are responsible for independently verifying every theorem, citation, implementation, empirical result, and disclosure in the final submission.

ETHICS STATEMENT

This work studies aggregation rules for language-model alignment, including harmlessness as one objective. Improvement in a game value is not an absolute safety certificate. Safety-critical deployment should retain hard constraints and independent safety evaluations in addition to the bargaining objective.

REPRODUCIBILITY STATEMENT

Algorithm 1 specifies the constrained training procedure. Appendix A provides the proximal-dual derivation and population convergence proof, while Appendices C, F, and H provide the matched experimental protocol, diagnostics, and judge details. A submission package should include anonymized code, exact prompts, random seeds, model checkpoints, opponent pools, dual traces, and the sampling budgets used for the regression and surplus estimates.

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A PROOFS

A.1 PROOF OF PROPOSITION 1

For a fixed comparator ν , define

$$\mathcal{J}_{k,\nu}(\pi) := J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)).$$

The first term is linear in π and the second does not depend on π , so $\mathcal{J}_{k,\nu}$ is affine. The pointwise minimum of affine functions is concave; hence V_{k,β_k} is concave. Since $|\Delta_k| \leq 1/2$,

$$|V_{k,\beta_k}(\pi) - V_{k,\beta_k}(\pi')| \leq \frac{1}{2} \mathbb{E}_{x \sim \mathcal{D}} \|\pi(\cdot | x) - \pi'(\cdot | x)\|_1,$$

which also proves continuity.

For $\beta_k > 0$, the inner minimization separates across prompts. Fix x and abbreviate $m(z) = m_k^\pi(z | x)$. The prompt-wise problem is

$$\min_{\nu \in \Delta(\mathcal{Y}_x)} \sum_z \nu(z) m(z) + \beta_k \sum_z \nu(z) \log \frac{\nu(z)}{\pi_{\text{ref}}(z | x)}.$$

The first-order condition under $\sum_z \nu(z) = 1$ gives

$$\nu(z) \propto \pi_{\text{ref}}(z | x) \exp\{-m(z)/\beta_k\},$$

which yields (8); substituting it back yields (9). The KL term makes the minimizer unique. Danskin's theorem therefore gives, for every feasible direction h satisfying $\sum_y h(y | x) = 0$,

$$DV_{k,\beta_k}(\pi)[h] = \mathbb{E}_{x \sim \mathcal{D}} \sum_y h(y | x) \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot | x)} \Delta_k(y, z | x).$$

Thus the coordinate derivative is

$$[\nabla_\pi V_{k,\beta_k}(\pi)]_{x,y} = \mathcal{D}(x) Q_k^\pi(y | x),$$

proving (10).

A.2 CONNECTION TO THE FIXED-REFERENCE

This subsection gives a finite-action illustration at the unregularized endpoint $\beta = 0$. It is not used in Assumption 1, the full-support analysis, or the finite-temperature experiments.

Define the fixed-reference improvement

$$a_k(\pi) = J_k(\pi, \pi_{\text{ref}}) = P_k(\pi \succ \pi_{\text{ref}}) - \frac{1}{2}. \quad (30)$$

Skew-symmetry gives $a_k(\pi_{\text{ref}}) = 0$.

Full cyclic example. For one prompt, let $\pi_{\text{ref}} = \delta_r$, where δ_y is the deterministic policy that always outputs y . Suppose A , B , and C each beat r with probability 0.75 and form the deterministic cycle

$$A \succ B, \quad B \succ C, \quad C \succ A.$$

Both

$$\pi_{\text{pure}} = \delta_A \quad \text{and} \quad \pi_{\text{mix}} = \frac{\delta_A + \delta_B + \delta_C}{3}$$

have the same fixed-reference improvement,

$$a(\pi_{\text{pure}}) = a(\pi_{\text{mix}}) = 0.75 - \frac{1}{2} = 0.25.$$

Their worst-opponent values are nevertheless different. For the pure policy, response C defeats A deterministically, so

$$V_0(\pi_{\text{pure}}) = \min_\nu J(\pi_{\text{pure}}, \nu) = J(\delta_A, \delta_C) = -\frac{1}{2}.$$

For the uniform mixture, consider first the pure opponent δ_A . The mixture outputs A , B , and C with equal probability. Against A , these responses respectively tie, lose, and win, giving centered payoffs 0, $-1/2$, and $+1/2$. Hence

$$J(\pi_{\text{mix}}, \delta_A) = \frac{1}{3} \left(0 - \frac{1}{2} + \frac{1}{2} \right) = 0.$$

By symmetry,

$$J(\pi_{\text{mix}}, \delta_B) = J(\pi_{\text{mix}}, \delta_C) = 0.$$

Against the reference,

$$J(\pi_{\text{mix}}, \delta_r) = 0.25,$$

because every response in the mixture beats r with probability 0.75. Since $J(\pi_{\text{mix}}, \nu)$ is linear in ν , no mixed opponent can achieve a value below the minimum over these pure opponents. Therefore

$$V_0(\pi_{\text{mix}}) = \min_{\nu} J(\pi_{\text{mix}}, \nu) = \min\{0, 0, 0, 0.25\} = 0.$$

Finally, the reference itself has

$$V_0(\pi_{\text{ref}}) = \min_{\nu} J(\delta_r, \nu) = -0.25,$$

since r loses to each of A , B , and C with probability 0.75. Thus

	$a(\pi)$	$V_0(\pi)$	$V_0(\pi) - V_0(\pi_{\text{ref}})$
π_{pure}	0.25	-0.50	-0.25
π_{mix}	0.25	0	0.25

The fixed reference therefore assigns the two policies the same score, even though the pure policy is fully exploitable by one response whereas the balanced mixture guarantees an average tie against every response in the cycle.

A.3 POLICY-INDUCED NASH BARGAINING AND CLASSICAL CONSEQUENCES

We prove Proposition 2. Since Π is compact and each V_{k, β_k} is continuous, $V_{\beta}(\pi)$ is bounded. The definition of $\mathcal{G}_{\beta}$ then implies boundedness. Closedness follows by taking any convergent sequence $v_n \in \mathcal{G}_{\beta}$, choosing policies π_n with $d \leq v_n \leq V_{\beta}(\pi_n)$, extracting a convergent policy subsequence, and applying continuity. Downward closure is immediate from the definition.

For convexity, take $v^1, v^2 \in \mathcal{G}_{\beta}$ with witnesses π_1, π_2 . Objective-wise concavity from Proposition 1 gives, for $\alpha \in [0, 1]$,

$$V_{\beta}(\alpha\pi_1 + (1 - \alpha)\pi_2) \geq \alpha V_{\beta}(\pi_1) + (1 - \alpha)V_{\beta}(\pi_2)$$

componentwise, so $\alpha v^1 + (1 - \alpha)v^2 \in \mathcal{G}_{\beta}$. Assumption 1 provides a point strictly above d , proving nondegeneracy.

For any $v \in \mathcal{G}_{\beta}$ with $v > d$, some policy π satisfies $v \leq V_{\beta}(\pi)$. Since the Nash objective is coordinatewise increasing,

$$\sum_k \log(v_k - d_k) \leq \sum_k \log(V_{k, \beta_k}(\pi) - d_k) = F(\pi).$$

Conversely, every $\pi \in \Pi_+$ induces a feasible value vector $V_{\beta}(\pi)$. Hence (12) realizes the classical Nash solution over $(\mathcal{G}_{\beta}, d)$.

The remaining statements are standard Nash-bargaining consequences, but we record the short arguments for completeness. Since the log objective tends to $-\infty$ when any surplus approaches zero, every optimizer lies strictly above d . Coordinatewise monotonicity implies Pareto optimality. Strict concavity of $\sum_k \log s_k$ over value vectors implies that all optimal policies induce the same surplus, hence the same game-value vector, although the policy itself can be nonunique.

Finally, under independent positive affine transformations

$$V'_k(\pi) = c_k V_k(\pi) + b_k, \quad d'_k = c_k d_k + b_k, \quad c_k > 0,$$

the transformed surplus is $s'_k(\pi) = c_k s_k(\pi)$ and

$$\sum_k \log s'_k(\pi) = \sum_k \log s_k(\pi) + \sum_k \log c_k.$$

The added term is independent of π , so the optimizer set is unchanged.

Table 5: Classical population guarantees of common aggregation rules under joint improvement. The Nash row follows from the standard bargaining solution; the other rows are supported by the finite counterexamples below.

rule	strict IR	Pareto optimal	unique value	affine invariant
positive-weight utilitarian	no	yes	no	no
absolute maxmin	no	no	no	no
surplus maxmin	yes	no	no	no
Nash bargaining	yes	yes	yes	yes

A.4 COUNTEREXAMPLES FOR ALTERNATIVE AGGREGATION RULES

The positive guarantees in Table 5 follow directly from the corresponding objectives. Because $w_k > 0$, the positive-weight utilitarian objective $\sum_k w_k V_k(\pi)$ is strictly increasing in every game-value coordinate, so none of its optimizers can be Pareto dominated. Under Assumption 1, surplus maxmin also attains a strictly positive minimum surplus because a jointly improving policy exists. The examples below establish the guarantees that can fail for the alternative rules.

All examples use three responses, so a policy is a probability vector $\pi = (\pi_1, \pi_2, \pi_3) \in \Delta_3$, and use the uniform reference $\pi_{\text{ref}} = (1/3, 1/3, 1/3)$. For objective k , Δ_k is the skew-symmetric pairwise payoff matrix: $(\Delta_k)_{ij}$ is the centered preference margin of response i against response j . At $\beta_k = 0$,

$$V_k(\pi) = \min_j (\pi^\top \Delta_k)_j, \quad d_k = V_k(\pi_{\text{ref}}), \quad s_k(\pi) = V_k(\pi) - d_k.$$

Thus $\pi^\top \Delta_k$ contains the expected payoff of π against each pure opponent, and $V_k(\pi)$ is its smallest coordinate.

Because each V_k is the minimum of linear functions, the optimizers below are obtained from small linear programs. For example, equal-weight utilitarian optimization can be written as

$$\max_{\pi, t_1, t_2} t_1 + t_2 \quad \text{s.t.} \quad t_k \leq (\pi^\top \Delta_k)_j \quad \forall k, j, \quad \pi \in \Delta_3.$$

Absolute maxmin uses one variable t with $t \leq (\pi^\top \Delta_k)_j$, while surplus maxmin uses $t + d_k \leq (\pi^\top \Delta_k)_j$ for every k, j .

Utilitarian failure of strict improvement. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & -1/2 \\ 1/2 & 0 & -1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & -1/2 \\ -1/4 & 1/2 & 0 \end{pmatrix}.$$

For the uniform reference,

$$\pi_{\text{ref}}^\top \Delta_1 = (1/3, 0, -1/3), \quad \pi_{\text{ref}}^\top \Delta_2 = (1/12, 0, -1/12),$$

and hence

$$d = (-1/3, -1/12).$$

First, joint improvement is feasible. The policy

$$\bar{\pi} = (1/4, 1/4, 1/2)$$

is only a feasibility witness; it gives

$$\bar{\pi}^\top \Delta_1 = (3/8, 1/8, -1/4), \quad \bar{\pi}^\top \Delta_2 = (0, 1/8, -1/16).$$

Therefore

$$V(\bar{\pi}) = (-1/4, -1/16), \quad s(\bar{\pi}) = V(\bar{\pi}) - d = (1/12, 1/48) > 0.$$

In contrast, solving the equal-weight utilitarian problem $\max_\pi V_1(\pi) + V_2(\pi)$ gives

$$\pi_U = (0, 1/5, 4/5).$$

For this policy,

$$\pi_U^\top \Delta_1 = (1/2, 2/5, -1/10), \quad \pi_U^\top \Delta_2 = (-1/10, 2/5, -1/10),$$

so

$$V(\pi_U) = (-1/10, -1/10), \quad s(\pi_U) = (7/30, -1/60).$$

Thus joint improvement is possible, yet a utilitarian optimum falls below the fallback on objective 2.

Absolute-maxmin failure of strict improvement. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & 1/2 \\ -1/4 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 1/4 & -1/2 \\ -1/4 & 0 & 1/2 \\ 1/2 & -1/2 & 0 \end{pmatrix}.$$

Again,

$$d = (-1/3, -1/12).$$

Joint improvement is feasible, as witnessed by

$$\bar{\pi} = (1/3, 7/15, 1/5),$$

for which

$$V(\bar{\pi}) = (-4/15, -1/60)$$

and therefore

$$s(\bar{\pi}) = (1/15, 1/15) > 0.$$

Absolute maxmin instead solves

$$\max_{\pi} \min\{V_1(\pi), V_2(\pi)\},$$

whose linear-program solution is

$$\pi_{AM} = (0, 4/5, 1/5).$$

For this policy,

$$\pi_{AM}^T \Delta_1 = (7/20, -1/10, 2/5), \quad \pi_{AM}^T \Delta_2 = (-1/10, -1/10, 2/5),$$

so

$$V(\pi_{AM}) = (-1/10, -1/10)$$

and

$$s(\pi_{AM}) = (7/30, -1/60).$$

Thus absolute maxmin maximizes the smallest *absolute* game value while allowing objective 2 to fall below its own fallback.

A Pareto-dominated surplus-maxmin optimum. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}.$$

The uniform reference gives

$$d = (-1/3, -1/6).$$

For objective 2, no policy can achieve $V_2(\pi) > 0$, while every mixture supported on the first two responses achieves $V_2(\pi) = 0$. Consequently,

$$s_2(\pi) \leq 1/6$$

for every policy, so the optimal surplus-maxmin value cannot exceed 1/6.

The two policies

$$\pi_c = (1/3, 2/3, 0), \quad \pi_e = (0, 1, 0)$$

satisfy

$$V(\pi_c) = (-1/6, 0), \quad s(\pi_c) = (1/6, 1/6),$$

and

$$V(\pi_e) = (0, 0), \quad s(\pi_e) = (1/3, 1/6).$$

Both therefore maximize

$$\min_k s_k(\pi)$$

at 1/6. However,

$$s(\pi_e) > s(\pi_c)$$

in the first coordinate and is equal in the second, so π_e Pareto dominates π_c . Thus the surplus-maxmin optimizer set can contain Pareto-dominated policies.

A.5 DUAL DERIVATION AND REGRESSION FIXED POINT

For multipliers $\lambda_k \geq 0$, the Lagrangian of (16) is

$$\mathcal{L}_t(\pi, u, \lambda) = \sum_k \log u_k + \sum_k \lambda_k (s_k(\pi) - u_k) - \eta_t^{-1} D(\pi \| \pi_t).$$

For fixed $\lambda_k > 0$, maximizing $\log u_k - \lambda_k u_k$ gives $u_k = 1/\lambda_k$ and value $-\log \lambda_k - 1$; if $\lambda_k = 0$, the maximum is unbounded. Substitution yields (17), and the remaining maximization over π is (18). Uniqueness of that maximizer and Danskin's theorem yield (19).

Assumption 1 provides strict feasibility: choose $\bar{\pi}$ with $s_k(\bar{\pi}) > 0$ and $0 < \bar{u}_k < s_k(\bar{\pi})$. Slater's condition therefore gives strong duality. Stationarity in u_k gives

$$\lambda_{t+1,k} = \frac{1}{u_{t+1,k}} > 0,$$

and complementary slackness then forces $u_{t+1,k} = s_k(\pi_{t+1})$. This proves Proposition 3.

Regression fixed point. Fix λ and let $\pi_t^*(\lambda)$ solve (18). First-order optimality gives, for every prompt and response pair in the support,

$$h_t(\pi_t^*(\lambda); x, y, y') = \eta_t \sum_k \lambda_k \left(Q_k^{\pi_t^*(\lambda)}(y | x) - Q_k^{\pi_t^*(\lambda)}(y' | x) \right). \quad (31)$$

At a fixed point of the alternating inner solve, the opponents in (26) are exactly $\nu_{k, \pi_t^*(\lambda)}^*$. Equation (25) therefore makes the conditional mean of the binary target equal to the right-hand side of (31). The squared-loss conditional-mean property implies that a population minimizer of (26) satisfies this pairwise identity. Full support of the pair proposal identifies all log-probability differences; normalization then recovers $\pi_t^*(\lambda)$.

A.6 PROOF OF THEOREM 1

Let $h(\pi) = \mathbb{E}_{x \sim \mathcal{D}} \sum_y \pi(y | x) \log \pi(y | x)$, so its Bregman divergence is $D(\pi \| \pi_t)$. Because the objective is increasing in u , the policy part of (16) is equivalently

$$\pi_{t+1} \in \arg \max_{\pi \in \Pi_+} \left\{ F(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (32)$$

The feasible set is nonempty by Assumption 1, and $F(\pi) \rightarrow -\infty$ as any surplus approaches zero. Thus the maximizer exists in Π_+ even when the center π_t is the reference $\pi_{\text{ref}} \notin \Pi_+$. Because π_t has full support and the remaining objective has finite directional derivatives at an interior surplus point, the inward derivative of $-D(\pi \| \pi_t)$ excludes zero-probability coordinates. Induction therefore gives a full-support, positive-surplus π_{t+1} .

Let g_{t+1} be a supergradient of F at π_{t+1} . First-order optimality of (32) and concavity of F imply, for every $\pi \in \Pi_+$,

$$F(\pi) - F(\pi_{t+1}) \leq \langle g_{t+1}, \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (33)$$

$$\leq \frac{1}{\eta_t} \langle \nabla h(\pi_{t+1}) - \nabla h(\pi_t), \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (34)$$

$$= \frac{1}{\eta_t} [D(\pi \| \pi_t) - D(\pi \| \pi_{t+1}) - D(\pi_{t+1} \| \pi_t)]. \quad (35)$$

For $t \geq 1$, π_t is feasible in the next subproblem. Comparing the optimum with π_t gives

$$F(\pi_{t+1}) - \frac{1}{\eta_t} D(\pi_{t+1} \| \pi_t) \geq F(\pi_t), \quad (36)$$

so the objective values are nondecreasing from π_1 onward.

Set $\pi = \pi^*$ in (35), multiply by η_t , and sum from $t = 0$ to $T - 1$:

$$\sum_{t=0}^{T-1} \eta_t (F(\pi^*) - F(\pi_{t+1})) \leq D(\pi^* \| \pi_{\text{ref}}). \quad (37)$$

Because the gaps are nonincreasing, the left-hand side is at least $(\sum_t \eta_t)(F(\pi^*) - F(\pi_T))$, which proves (28).

Corollary 1 (Convergence of the surplus profile). *Under the conditions of Theorem 1, let $s^* = s(\pi^*)$ denote the unique surplus vector induced by a Nash-optimal policy. Then*

$$\|s(\pi_T) - s^*\|_2^2 \leq \frac{2D(\pi^* \|\pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (38)$$

Hence a constant proximal step gives $\|s(\pi_T) - s^*\|_2 = O(T^{-1/2})$.

Proof. This is a stronger consequence of Nash-welfare convergence rather than a separate requirement for the NBPO update. For every objective, $V_k(\pi) \in [-1/2, 1/2]$, and $d_k = V_k(\pi_{\text{ref}}) \leq 0$ because $\nu = \pi_{\text{ref}}$ is feasible and $J_k(\pi_{\text{ref}}, \pi_{\text{ref}}) = 0$. Hence every positive surplus is at most one. On $(0, 1]^K$, the function $G(s) = \sum_k \log s_k$ is 1-strongly concave in Euclidean norm. Proposition 2 implies that all Nash-optimal policies induce the same game-value vector, and therefore the same surplus vector s^* . First-order optimality over the convex comprehensive surplus set induced by (14) gives

$$G(s^*) - G(s) \geq \frac{1}{2} \|s - s^*\|_2^2. \quad (39)$$

Applying this inequality to $s = s(\pi_T)$ and using (28) proves (38). $\square$

B OBJECTIVE-WISE EVALUATION AND GAME-SURPLUS DIAGNOSTICS

Table 6 is the fill-ready objective-wise counterpart of the primary external comparison in Table 2. All entries must use the same independent evaluator and prompt split. Table 7 retains the currently available adaptive game-value surpluses only as a diagnostic of the NBPO-aligned objective; because this metric is defined by NBPO’s game construction, it should not be the sole headline ranking of unrelated algorithms.

Table 7 gives the objective-wise decomposition of the existing game-surplus results. These values are useful for diagnosing which objectives each policy sacrifices, but they are defined by the adaptive game evaluation introduced by NBPO and are therefore secondary to the common external metric in Table 6.

C IMPLEMENTATION AND HYPERPARAMETERS

Following prior preference-optimization and multi-objective alignment work, we report response generation, preference scoring, policy optimization, method-specific loop parameters, and compute separately (Munos et al., 2024; Zhang et al., 2025; Zhong et al., 2024b; Zhang et al., 2026). Tables 8 and 9 are fill-ready records for the final experiments. Values already stated in the manuscript are prefilled; empty cells should be completed from the released configurations rather than inferred.

Implementation audit. The entries above distinguish the released configuration from the final protocol we recommend reporting. The current validation split is pair-level; final experiments should split prompts before response generation so that a prompt or response cannot appear in both training and validation. The released implementation also records $R = 1$: it performs one opponent reweighting per dual update rather than explicitly iterating the policy–opponent fixed point to convergence. This approximation should either be exposed as a tunable R with a reported fixed-point residual or stated explicitly as the implemented approximation to Section 5.2. Finally, the reported query budgets mix semantic comparisons and order-swapped judge calls across panels; the final artifact should fill the two separate accounting rows above.

D REWARD-MODEL BASELINE PROTOCOL

Reward-model baselines require additional choices that can materially affect the comparison. Table 10 must be completed before the corresponding rows in Tables 1 and 2 are interpreted.

Table 6: Pending objective-wise held-out win rates W_k^{eval} against the common reference. Blank cells denote required prompt-disjoint, multi-seed evaluations.

method	obj. 1	obj. 2	obj. 3	obj. 4	average
<i>ULTRAFEEDBACK: instruction following, truthfulness, honesty, helpfulness</i>					
Base = π_{ref}	0.500	0.500	0.500	0.500	0.500
NBPO	0.528	0.524	0.539	0.547	0.535
PROSPER/MaxEntBW	0.518	0.496	0.544	0.605	0.541
Game-utilitarian	0.479	0.467	0.504	0.550	0.500
Game-absolute-maxmin	0.472	0.459	0.501	0.565	0.499
Game-surplus-maxmin	0.502	0.517	0.574	0.603	0.549
Uniform-INPO	0.540	0.556	0.608	0.617	0.580
MOPO	0.507	0.508	0.512	0.516	0.511
MaxMin-RLHF	0.554	0.537	0.567	0.569	0.557
Rewarded Soups	0.559	0.534	0.553	0.582	0.557
Panacea	0.568	0.554	0.570	0.594	0.572
<i>TL;DR: coverage, faithfulness, conciseness, helpfulness</i>					
Base = π_{ref}	0.500	0.500	0.500	0.500	0.500
NBPO	0.572	0.593	0.528	0.539	0.558
PROSPER/MaxEntBW	0.673	0.590	0.444	0.661	0.592
Game-utilitarian	0.692	0.599	0.415	0.675	0.595
Game-absolute-maxmin	0.694	0.493	0.365	0.675	0.557
Game-surplus-maxmin	0.578	0.727	0.613	0.491	0.602
Uniform-INPO	0.399	0.668	0.780	0.228	0.519
MOPO	0.574	0.557	0.513	0.540	0.546
MaxMin-RLHF	0.710	0.517	0.346	0.682	0.564
Rewarded Soups	0.571	0.618	0.580	0.506	0.569
Panacea	0.649	0.616	0.584	0.560	0.602
<i>SAFERLHF: helpfulness, harmlessness</i>					
Base = π_{ref}	0.500	0.500	–	–	0.500
NBPO	0.740	0.696	–	–	0.718
PROSPER/MaxEntBW	0.794	0.760	–	–	0.777
Game-utilitarian	0.794	0.780	–	–	0.787
Game-absolute-maxmin	0.781	0.784	–	–	0.782
Game-surplus-maxmin	0.778	0.781	–	–	0.780
Uniform-INPO	0.447	0.495	–	–	0.471
MOPO	0.740	0.720	–	–	0.730
MaxMin-RLHF	0.768	0.705	–	–	0.736
Rewarded Soups	0.755	0.728	–	–	0.742
Panacea	0.750	0.703	–	–	0.726

E ADDITIONAL EXTERNAL EVALUATION

The general-capability table reserves IFEval and TruthfulQA for their official task-specific scoring. Because safety uses a different metric and direction, HarmBench is reported separately in Table 11 rather than mixed into Table 1.

F OPTIMIZATION DIAGNOSTICS AND ROBUSTNESS

Table 12 separates the numerical dual solve from the harder neural-realization problem. The published dual is already stable: re-solving the same frozen problem with $M = 4,000, 40,000$, or $400,000$ changes the multipliers only at small numerical precision, whereas the learned policy remains sensitive to the opponent temperature. On the current ULTRAFEEDBACK panel, decreasing β from 0.25 to 0.15 reduces the minimum surplus from 0.028 to 0.014, while the fixed-reference limit $\beta \rightarrow \infty$ has the higher point estimate (0.040). The remaining rows should therefore prioritize the sources of stochastic and fitting error—comparator-pool size, learner comparisons per comparator, response-pair sampling, and the policy-opponent fixed-point approximation—rather than extending the already converged dual budget.

G DIAGNOSING THE ADAPTIVE-GAME GAP

Table 3 shows that the simpler fixed-reference Nash variant currently has a slightly better point estimate than adaptive-game NBPO. The following tests separate three explanations: the adaptive population objective may not help on this pool, the neural regression may fail to realize its target, or the pool may contain too little intransitivity for an adaptive opponent to matter. All diagnostics

Table 7: Secondary adaptive game-value surplus diagnostic for the existing single-run checkpoints. This NBPO-aligned metric is retained for mechanism analysis rather than used as the primary cross-method ranking.

method	obj. 1	obj. 2	obj. 3	obj. 4	min <i>s</i>	avg <i>s</i>
ULTRAFEEDBACK on Llama-3.1-8B-Instruct: instruction following, truthfulness, honesty, helpfulness						
NBPO (ours)	0.039	0.049	0.055	0.109	0.039	0.063
PROSPER/MaxEntBW	-0.035	-0.051	-0.046	0.068	-0.051	-0.016
Utilitarian	-0.064	-0.077	-0.070	0.027	-0.077	-0.046
Absolute maxmin	-0.043	-0.062	-0.051	0.059	-0.062	-0.024
Surplus maxmin	-0.014	-0.000	0.020	0.108	-0.014	0.028
Uniform-INPO	-0.008	0.022	0.020	0.079	-0.008	0.028
MOPO	0.031	0.026	0.022	0.032	0.022	0.028
HT-MNPO (helpfulness)	-0.057	-0.073	-0.067	0.050	-0.073	-0.037
HT-MNPO (truthfulness)	-0.050	-0.049	-0.041	0.023	-0.050	-0.029
HT-MNPO (honesty)	0.022	0.027	0.025	0.075	0.022	0.037
HT-MNPO (instr. following)	0.008	0.026	0.032	0.065	0.008	0.033
ULTRAFEEDBACK on Qwen3-8B: helpfulness, truthfulness, honesty, instruction following						
NBPO (ours)	0.072	0.018	0.043	0.031	0.018	0.041
PROSPER/MaxEntBW	0.039	-0.002	0.016	0.017	-0.002	0.018
Utilitarian	0.038	0.012	0.016	0.012	0.012	0.019
Absolute maxmin	0.041	0.005	0.018	0.012	0.005	0.019
Surplus maxmin	0.038	0.004	0.014	0.009	0.004	0.016
Uniform-INPO	-0.005	-0.001	-0.001	-0.003	-0.005	-0.002
MOPO	-0.002	-0.001	-0.005	0.001	-0.005	-0.001
HT-MNPO (helpfulness)	-0.002	-0.001	-0.001	0.002	-0.002	-0.001
HT-MNPO (truthfulness)	0.017	0.001	0.016	0.014	0.001	0.012
HT-MNPO (honesty)	0.000	0.005	0.001	-0.003	-0.003	0.001
HT-MNPO (instruction following)	-0.004	0.004	0.001	0.002	-0.004	0.001
TL;DR on Llama-3.1-8B-Instruct: coverage, faithfulness, conciseness, helpfulness						
NBPO (ours)	0.198	0.079	-0.001	0.178	-0.001	0.114
PROSPER/MaxEntBW	0.205	0.031	-0.068	0.175	-0.068	0.086
Utilitarian	0.221	0.072	-0.046	0.188	-0.046	0.109
Absolute maxmin	0.213	0.005	-0.079	0.182	-0.079	0.080
Surplus maxmin	0.081	0.090	0.018	0.066	0.018	0.064
Uniform-INPO	-0.051	-0.052	-0.011	-0.050	-0.052	-0.041
MOPO	0.061	0.009	-0.001	0.052	-0.001	0.030
HT-MNPO (coverage)	0.051	-0.010	-0.007	0.045	-0.010	0.020
HT-MNPO (faithfulness)	0.074	-0.020	-0.026	0.060	-0.026	0.022
HT-MNPO (conciseness)	0.126	-0.037	-0.079	0.106	-0.079	0.029
HT-MNPO (helpfulness)	0.037	-0.025	-0.035	0.048	-0.035	0.006
SAFERLHF on Llama-3.1-8B-Instruct: helpfulness, harmlessness						
NBPO (ours)	0.118	0.101	—	—	0.101	0.110
PROSPER/MaxEntBW	0.151	0.116	—	—	0.116	0.134
Utilitarian	0.157	0.138	—	—	0.138	0.148
Absolute maxmin	0.154	0.144	—	—	0.144	0.149
Surplus maxmin	0.150	0.143	—	—	0.143	0.147
Uniform-INPO	-0.077	-0.055	—	—	-0.077	-0.066
MOPO	0.102	0.096	—	—	0.096	0.099
HT-MNPO (helpfulness)	-0.024	-0.049	—	—	-0.049	-0.036
HT-MNPO (harmlessness)	-0.023	-0.050	—	—	-0.050	-0.037

should use the same audited ULTRAFEEDBACK prompt intersection and the same frozen preference matrices whenever possible.

H JUDGE INSTANTIATION AND PREFERENCE LABELING

Training judges. Use the same prompted pairwise interface on both panels. For each objective, Qwen3-32B compares two responses under an attribute-specific rubric. Each pair is queried in both presentation orders, and the verdicts are swap-averaged. The fixed-reference precursor also contains deterministic Beaver reward and cost heads; those runs should not be conflated with the prompted-model main comparison.

Evaluation judges. Evaluation is separate from training and uses Phi-4 and Llama-3.3-70B. Each evaluator returns $[A]$, $[B]$, or $[TIE]$, scored as 1, 0, or 1/2, in both presentation orders. This makes the empirical evaluation model skew-symmetric after averaging.

Rubrics. Helpfulness asks which response better accomplishes the user’s stated goal; correctness asks which is more accurate; coherence asks which is clearer and better organized; conciseness asks which conveys the needed content with less padding; and harmlessness asks which is safer. The judge is instructed to evaluate only the designated objective.

Prompt template. For concreteness, the helpfulness system message and pairwise user message are

```
System: You compare two AI responses on HELPFULNESS ONLY:
which better helps the user accomplish their stated goal
(relevance, substance, actionable detail). Ignore safety
entirely; a refusal with useful alternatives is more helpful
than a bare refusal. End with exactly [[A]] or [[B]] or
[[TIE]].
User: [User prompt] {prompt} [Response A] {A} [Response B]
{B} Verdict:
```

The other objectives use the same template with only the rubric changed, and each pair is queried again after swapping responses A and B.

From verdicts to regularized opponents. For a current policy π_t and a reference comparator z , estimate

$$\hat{m}_k^{\pi_t}(z | x) = \frac{1}{L} \sum_{\ell=1}^L (B_k(y_\ell, z) - \frac{1}{2}), \quad y_\ell \sim \pi_t(\cdot | x).$$

Reference responses are reweighted by $\exp\{-\hat{m}_k^{\pi_t}(z | x)/\beta_k\}$ and normalized to sample $z_k \sim \hat{\nu}_{k, \pi_t}^*$. For a policy pair (y, y') sharing z_k , the binary difference in (24) enters the regression target directly. No scalar reward model or value function is fitted by NBPO.

Table 8: Data, sampling, evaluation, and NBPO-specific settings. Empty cells are fill-ready placeholders.

setting	ULTRAFEEDBACK/Llama	ULTRAFEEDBACK/Qwen	TL;DR/Llama	SAFERLHF/Llama
Objective panel	instr. following; truth-helpfulness; truthfulness; coverage; faithfulness; helpfulness; harmless-fulness; honesty; helpful-honesty; instr. following conciseness; helpfulness ness			
Base/reference policy	Llama-3.1-8B-Instruct	Qwen3-8B	Llama-3.1-8B-Instruct	Llama-3.1-8B-Instruct
Training preference judge	Qwen3-32B	Qwen3-32B	Qwen3-32B	Qwen3-32B
Primary evaluation judge	Llama-3.3-70B-Instruct	Llama-3.3-70B-Instruct	Llama-3.3-70B-Instruct	Llama-3.3-70B-Instruct
Secondary evaluation judge	Phi-4 (secondary check)	Phi-4 (secondary check)	Phi-4 (secondary check)	Phi-4 (secondary check)
Qwen thinking mode	–	reasoning trace suppressed; record exact flag	–	–
Training prompts	7,000	300	600	600
Validation split	current: 10% pair-level; final rerun: prompt-level	current: 10% pair-level; final rerun: prompt-level	current: 10% pair-level; final rerun: prompt-level	current: 10% pair-level; final rerun: prompt-level
Held-out evaluation prompts	657	100	100	100
Outer proximal stages T	1	1	1	1
Dual iterations per stage M	40,000	4,000	300,000	300,000
Fixed-point iterations per dual step 1	–	1	1	1
R	–	–	–	–
Proximal coefficient η_t	1.0	1.0	1.0	1.0
Dual step size γ_m	0.5	0.05	2.0	2.0
Opponent temperatures $\beta_{1:K}$	0.25 (all k)	0.25 (all k)	0.25 (all k)	0.25 (all k)
Dual box $[\lambda_{\min}, \lambda_{\max}]$	$[10^{-3}, 10^3]$	$[10^{-3}, 10^3]$	$[10^{-3}, 10^3]$	$[10^{-3}, 10^3]$
Dual initialization/warm start	$\lambda^{(0)} = 1$, no warm start	$\lambda^{(0)} = 1$, no warm start	$\lambda^{(0)} = 1$, no warm start	$\lambda^{(0)} = 1$, no warm start
Dual stopping criterion and tolerance	fixed budget M ; KKT residual $< 4 \times 10^{-15}$	fixed budget M ; KKT residual 6.6×10^{-3}	fixed budget M	fixed budget M
Policy responses per prompt	4	4	4	4
Opponent candidate pool per prompt	4	4	4	4
Reference comparators per prompt	4	4	4	4
Learner comparisons per comparator L	4	4	4	4
Response-pair proposal ξ	all 6 unordered pairs	all 6 unordered pairs	all 6 unordered pairs	all 6 unordered pairs
Presentation orders per judged pair	2	2	2	2
Judge repetitions per order	1	1	1	1
Training response temperature	0.9	0.9	0.9	0.9
Training response top- p /top- k	0.95 / –	0.95 / –	0.95 / –	0.95 / –
Maximum training response tokens	512	512	512	512
Training-judge temperature	0.0 (greedy)	0.0 (greedy)	0.0 (greedy)	0.0 (greedy)
Training-judge top- p /top- k	– / –	– / –	– / –	– / –
Maximum judge tokens	512	512	512	512
Evaluation generation temperature	0.7	0.7	0.7	0.7
Evaluation generation top- p /top- k	0.9 / –	0.9 / –	0.9 / –	0.9 / –
Maximum evaluation response tokens	1,024	1,024	1,024	1,024
IFEval protocol/version	lm-eval 0.4.9, task ifeval v4.0, 0-shot, greedy, 541 prompts; prompt-level strict acc.	lm-eval 0.4.9, task ifeval v4.0, 0-shot, greedy, 541 prompts; prompt-level strict acc.	lm-eval 0.4.9, task ifeval v4.0, 0-shot, greedy, 541 prompts; prompt-level strict acc.	lm-eval 0.4.9, task ifeval v4.0, 0-shot, greedy, 541 prompts; prompt-level strict acc.
TruthfulQA protocol/version	lm-eval 0.4.9, truthfulqa_mc2 v3.0, 0-shot, 817 questions	lm-eval 0.4.9, truthfulqa_mc2 v3.0, 0-shot, 817 questions	lm-eval 0.4.9, truthfulqa_mc2 v3.0, 0-shot, 817 questions	lm-eval 0.4.9, truthfulqa_mc2 v3.0, 0-shot, 817 questions
HarmBench protocol/version	official walledai/HarmBench 300 standard+contextual behaviours, DirectRequest, greedy, ≤ 512 new tokens, HarmBench-Llama-2-13b-c1s	–	–	–
Held-out monitoring batch size	100 prompts (prompt-disjoint)	–	–	100 prompts (prompt-disjoint)
Reported judged-comparison budget	448,000	28,725	38,363	38,400
Unique semantic comparisons	448,000 ($4 \times 7,000 \times 4 \times 4$)	not reconstructible (ten-37,248 ($4 \times 582 \times 4 \times 4$))	not reconstructible (ten-74,496)	19,200 ($2 \times 600 \times 4 \times 4$)
Order-swapped judge calls (incl. retries)	896,000	not reconstructible (ten-74,496)	–	38,400
Scale-test vectors	$c^{(A)}$ = (0.2, 0.6, 1.0, 1.7); $c^{(B)}$ = (3.0, 0.5, 1.0, 0.25); $c^{(C)}$ = (1.0, 1.0, 1.0, 5.0)	–	–	–
Scale-test solver tolerance	surplus floor 5×10^{-3} ; KKT residual $< 4 \times 10^{-15}$	–	–	–

Table 9: Policy optimization, model adaptation, tuning, and compute settings. Empty cells are fill-ready placeholders.

setting	ULTRAFEEDBACK/Llama		ULTRAFEEDBACK/Qwen TL;DR/Llama		SAFERLHF/Llama	
Adaptation mode (full/PEFT)	full fine-tuning		full fine-tuning		full fine-tuning	
Trainable modules	all transformer	parameters	all transformer	parameters	all transformer	parameters
Optimizer	Adafactor		Adafactor		Adafactor	
Learning rate	5×10^{-7}		5×10^{-7}		5×10^{-7}	
Learning-rate scheduler	cosine		cosine		cosine	
Warmup ratio/steps	0.1 / -		0.1 / -		0.1 / -	
Weight decay	0		0		0	
Per-device batch size	2		2		2	
Gradient accumulation steps	8		8		8	
Global batch size	16		16		16	
Epochs or optimizer updates per stage	300 updates		300 updates		300 updates	
Maximum input length	1,024		1,024		1,024	
Maximum sequence length	2,048		2,048		2,048	
Gradient clipping	1.0 (framework default)		1.0 (framework default)		1.0 (framework default)	
Numerical precision	bf16		bf16		bf16	
Gradient checkpointing	yes (non-reentrant)		yes (non-reentrant)		yes (non-reentrant)	
Distributed-training strategy	single device (no sharding)		single device (no sharding)		single device (no sharding)	
Random seeds	42		42		42	
Final independent training seeds			at least 3 per reported main comparison; pending			
Hyperparameters tuned and candidate grid	$\eta_t \in \{1.0, \text{pooled ent.}\}; \text{budget} \in \{300, 600\} \text{ only}; \beta_k \in \{0.15, 0.25, 0.5, 1, \infty\}$ (solver side)		$\eta_t \in \{1.0, \text{pooled ent.}\}; \text{budget} \in \{300, 600\} \text{ only}$		$\eta_t \in \{1.0, \text{pooled ent.}\}; \text{budget} \in \{300, 600\} \text{ only}$	
Validation set and selection metric	prompt-disjoint held-out pairs of the same pair set; lower held-out normalized MSE of (26)		prompt-disjoint held-out pairs of the same pair set; lower held-out normalized MSE of (26)		prompt-disjoint held-out pairs of the same pair set; lower held-out normalized MSE of (26)	
Number of tuning trials	9		2		2	
Hardware and number of accelerators	$1 \times \text{H200 (train)}; 4 \times \text{H200 (train)}; 4 \times \text{H200 (decode/judge)}$		$1 \times \text{H200 (train)}; 4 \times \text{H200 (train)}; 4 \times \text{H200 (decode/judge)}$		$1 \times \text{H200 (train)}; 4 \times \text{H200 (train)}; 4 \times \text{H200 (decode/judge)}$	
Wall-clock time/GPU-hours per run	0.85 H200-h train (300 upd.; 1.64 at 600); 0.65 H200-h evaluation judg-ing per arm		0.85 H200-h train; 0.25 H200-h evaluation judg-ing per arm		0.85 H200-h train; 0.29 H200-h evaluation judg-ing per arm	

Table 10: Fill-ready implementation record for reward-model-based multi-objective baselines.

method	reward model(s) and training data	policy optimizer and KL coefficient	trade-off/selection hyper-parameters	hyper- selected operating point and compute
MaxMin-RLHF	ArmoRM-Llama3-8B-v0.1 objective heads scored on the panel's own shared response pool; the panel's objectives are the known groups, so Algorithm 2's EM over latent user clusters is not needed	DPO on the worst objective's reward-ranked pairs, size: $\beta = 0.1$, reference fixed at π_{ref} , Adafactor, lr 5×10^{-7} , 3×100 updates	adversarial mixture/step	instr./honesty/truth. selected on UF/Llama, cover./concise/help. on TL;DR, help./harm./help. on SAFERLHF; 2×300 validation samples per round; ≈ 1.3 H200-hours per panel
Rewarded Soups	same reward heads; one expert per objective, each from the panel's own shared initialization	DPO per expert, $\beta = 0.1$, 300 updates each, then linear weight interpolation	interpolation grid and validation rule:	$\lambda = (.55, .15, .15, .15)$ UF/Llama, $(.15, .15, .15, .55)$ UF/Qwen, uniform TL;DR, $(.5, .5)$ SAFERLHF; 298/100/100/100 validation prompts; ≈ 3.3 H200-hours per 4-objective panel
Panacea	same reward heads; the objective DPO objectives are aggregated online rather than pre-mixed	SVD-LoRA, $k = 8$ preference-conditioning grid and validation rule: singular values, $m = K$ injected as the remaining singular values with a learnable per-matrix scale, AdaLoRA orthogonality penalty, LS aggregation, $\beta = 0.1$, lr 2×10^{-4} , 300 updates		$\lambda = (.15, .15, .55, .15)$ UF/Llama, $(.15, .55, .15, .15)$ UF/Qwen, $(.15, .15, .55, .15)$ TL;DR, $(.4, .6)$ SAFERLHF; ≈ 1.2 H200-hours per panel

Table 11: Planned safety-transfer evaluation on HarmBench. Report the official attack-success rate (ASR; lower is better) using the same frozen checkpoints as the main comparison. Blank cells denote pending runs.

method	HarmBench ASR ↓
Base = $\pi_{\text{ref}} = \pi_0$	0.223
NBPO (ours)	0.290
PROSPER/MaxEntBW	0.310
Utilitarian	0.350
Absolute maxmin	0.433
Surplus maxmin	0.320
Uniform-INPO	0.377
MOPO	0.240

Table 12: Optimization sensitivity on the 698-prompt ULTRA FEEDBACK panel. Existing matched measurements are filled; blank cells denote pending runs. Opponent entropy is measured over the four reference comparators, fixed-point residual measures policy–opponent consistency, and dual residual is $\|\hat{s} - \lambda^{-1}\|_\infty$.

ablation	setting	min s	avg s	$F(\pi)$	opp. entropy	fixed-point res.	dual res. reg. MSE
Default configuration	$\beta = 0.25, M = 40,000$	0.028	0.056	−11.82	1.160	-3.6×10^{-15}	73.9
Opponent temperature β	$\beta = 0.15$	0.014	0.046	−13.07	0.956	-3.6×10^{-15}	74.1
Opponent temperature β	$\beta = 0.50$	0.0665	0.0736	−10.45	1.315	-3.6×10^{-15}	
Opponent temperature β	$\beta = 1.00$	0.0672	0.0742	−10.41	1.367	-3.6×10^{-7}	
Opponent temperature β	$\beta \rightarrow \infty$ (fixed reference)	0.040	0.061	−11.40	1.386	—	—
Fixed-point budget R	$R = 1$, no damp.	0.0658	0.0730	−10.48	1.160	2.0×10^{-9}	5.0×10^{-7}
Fixed-point budget R	$R = 64$, damp. 0.9	0.0658	0.0730	−10.48	1.160	2.1×10^{-13}	5.0×10^{-7}
Dual budget M	$M = 4,000$	0.0658	0.0730	−10.48	1.160	-2.0×10^{-5}	
Dual budget M	$M = 400,000$				1.160	-3.6×10^{-15}	
Reference comparator count	2 of 4	0.0717	0.0798	−10.13	0.569	2.6×10^{-11}	6.5×10^{-8}
Learner comparisons per comparator L	$L = 2$ of 4	0.0472	0.0517	−11.86	1.124	3.1×10^{-8}	6.2×10^{-5}
Response-pair sample count	3 of 6 pairs	—	—	—	—	—	—

Table 13: Population-target versus neural-realization diagnostic. Solve each Nash objective exactly on the frozen finite response pool, then evaluate both the exact finite-pool solution and the trained neural checkpoint with the same adaptive game-value metric. Blank cells denote pending exact solves or measurements.

representation	exact min s	exact avg s	trained min s	trained avg s	min- s realization gap	avg- s realization gap
Adaptive game value	0.0658	0.0730	0.028	0.056	0.038	0.017
Fixed-reference margin	0.0679	0.0749	0.040	0.061	0.028	0.014

Table 14: Regression-realization diagnostics for the pairwise target in (26). Evaluate on both training and prompt-disjoint held-out pairs. The zero-predictor MSE is $\mathbb{E}[Y^2]$ for target $Y = \eta_t \sum_k \lambda_k Z_{t,k}$. Blank cells denote pending measurements.

representation	split	target RMS	zero-pred. MSE	reg. MSE	normalized MSE	R^2 / corr.	sign agreement
Adaptive game value	train	8.56	73.3	73.9	1.008	0.246	0.581
Adaptive game value	held-out	9.74	94.96	94.75	0.9977	0.148	0.558
Fixed-reference margin	train	8.70	75.73	78.80	1.0405	0.235	0.579
Fixed-reference margin	held-out	8.66	75.01	76.43	1.0189	0.119	0.547

Table 15: Cycle-conditioned comparison of adaptive-game and fixed-reference Nash policies. Define the high-intransitivity subset from the frozen pairwise preference matrices before evaluating policies. Blank cells denote pending analyses.

prompt subset	prompts	cycle / intransitivity rate	adaptive min s	fixed-ref. min s	adaptive — fixed	ranking disagreement
All audited prompts	698	0.565	0.028	0.040	−0.012	0.306
Low-intransitivity subset	3597	0.259	0.0815	0.0838	−0.0023	0.114
High-intransitivity subset	3403	0.674	0.0459	0.0477	−0.0018	0.366

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Table 16: Existing NBPO-aligned component diagnostic on 698 ULTRAFEEDBACK prompts. These game-surplus values are secondary to the independent metrics in the main text.

variant	min s	avg. s	$F(\pi)$	avg. ref. win
NBPO	0.028	0.056	-11.82	0.556
Fixed-reference NBPO	0.040	0.061	-11.40	0.561
Game-utilitarian	-0.073	-0.048	-	0.452
Game-surplus-maxmin	-0.044	-0.006	-	0.494